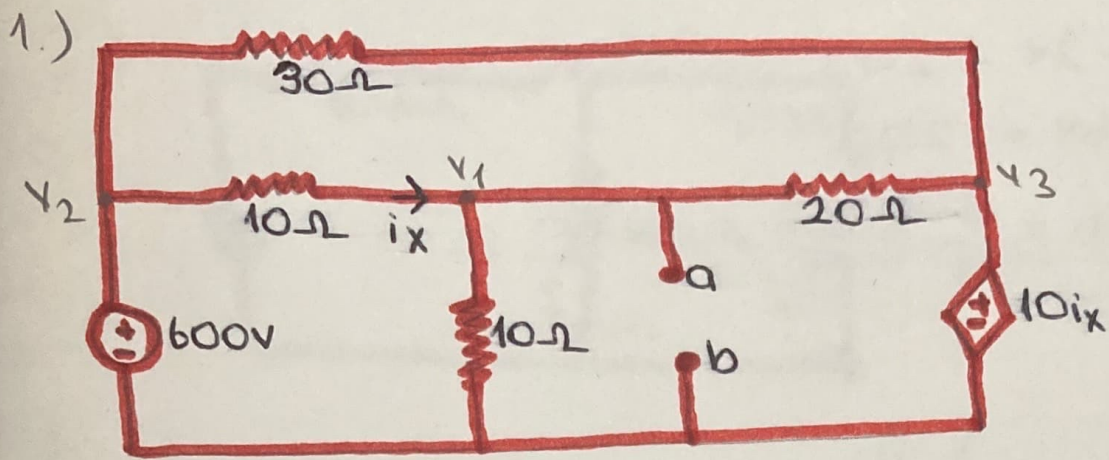




$$V_2 = 600V$$

$$V_3 = 10i_x$$

$$V_1 = V_{Th}$$

$$i_x = \frac{V_2 - V_1}{10} = \frac{600 - V_1}{10} \rightarrow 10i_x = 600 - V_1$$

For node - 1:

$$\frac{V_1 - 600}{10} + \frac{V_1}{10} + \frac{V_1 - 10i_x}{20} = 0$$

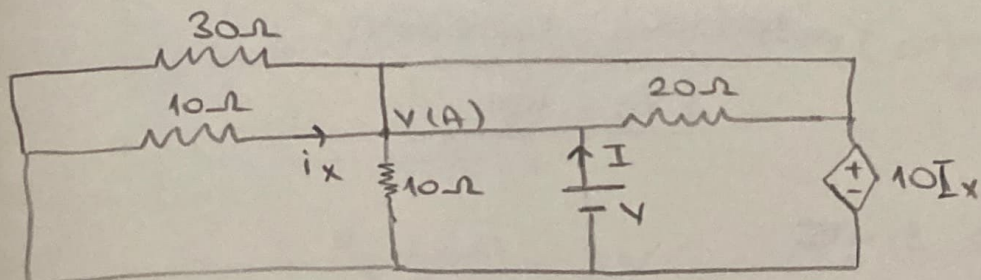
$$= 2V_1 - 1200 + 2V_1 + V_1 - 10\left(\frac{600 - V_1}{10}\right) = 0$$

$$= 5V_1 - 1200 - (600 - V_1) = 0$$

$$= 6V_1 - 1800 = 0$$

$$6V_1 = 1800 \rightarrow \underline{V_1 = 300V} = V_{Th}$$

↳ deactivate independent sources



$$i_x = \frac{0 - V}{10} = -\frac{V}{10}$$

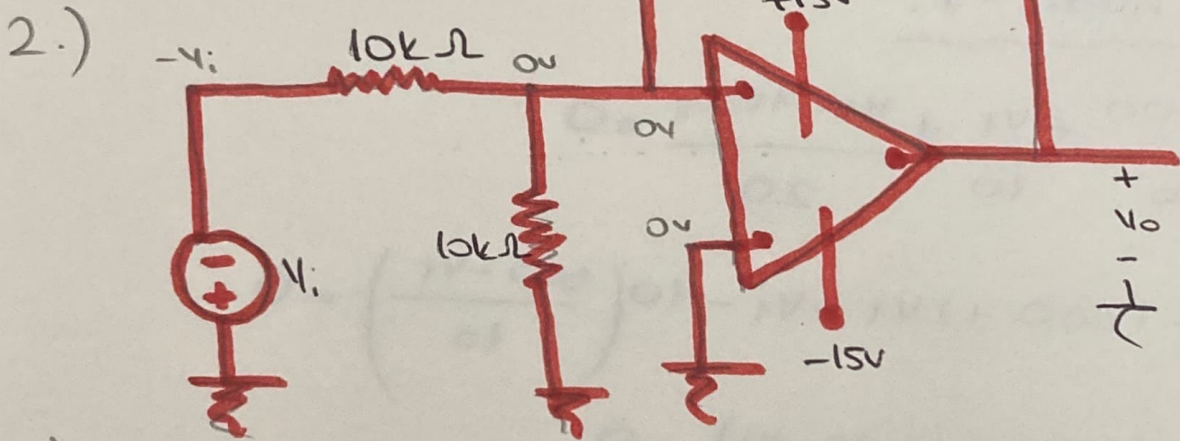
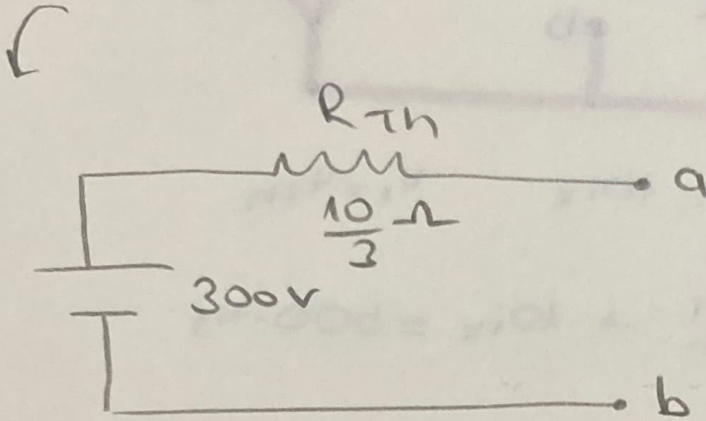
For node - A:

$$\frac{V}{10} + \frac{V}{10} + \frac{V - 10I_x}{20} = I$$

$$2V + 2V + 2V = 20I$$

$$6V = 20I$$

$$R_{Th} = \frac{V}{I} = \frac{20}{6} \Omega = \frac{10}{3} \Omega$$



a.)

$$\frac{-V_i - 0}{10000} + \frac{0 - 0}{1000} = 0 + \frac{0 - V_o}{100000}$$

$$\frac{-V_i}{10000} = \frac{-V_o}{100000}$$

$$\rightarrow 10000V_o = 100000V_i$$

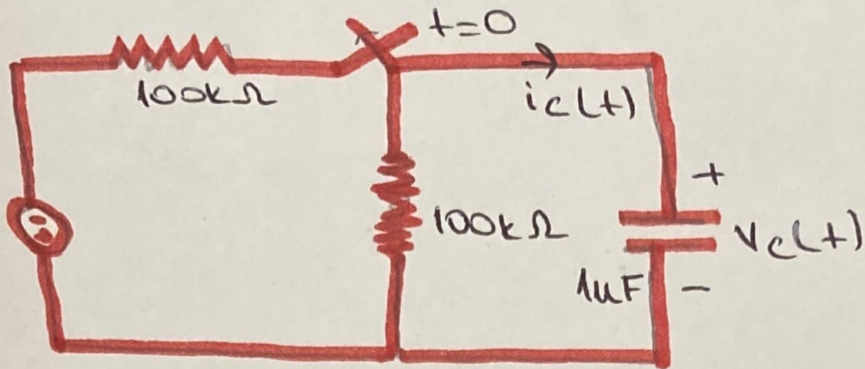
$$V_o = 10V_i$$

b.) $-15 \leq V_o \leq +15$

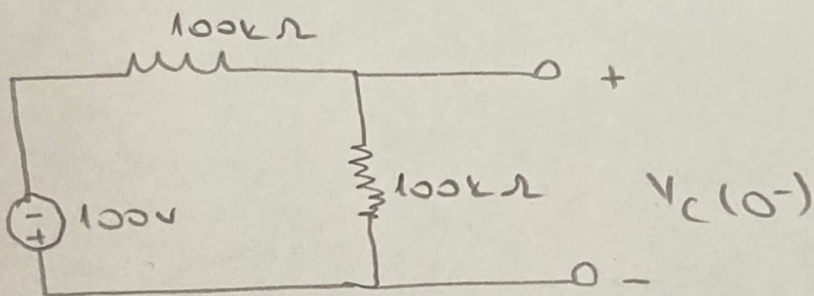
$$-15 \leq 10V_i \leq +15$$

$$\boxed{-1.5 \leq V_i \leq 1.5}$$

3.)



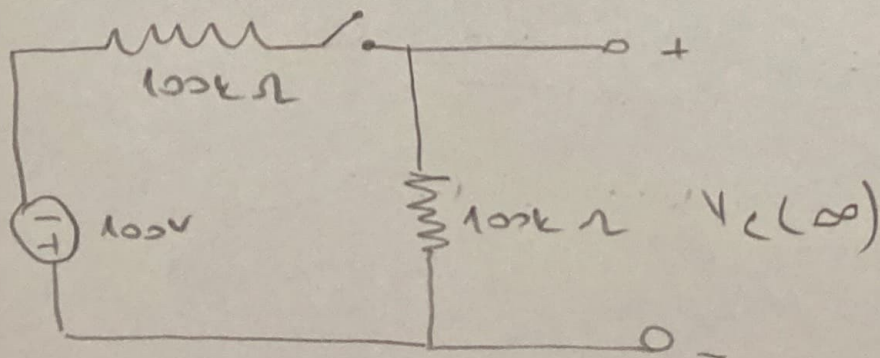
At $t=0^-$



$$V_c(0^-) = \left(\frac{100}{100+100} \right) \times (-100)$$

$$V_c(0^-) = -50V$$

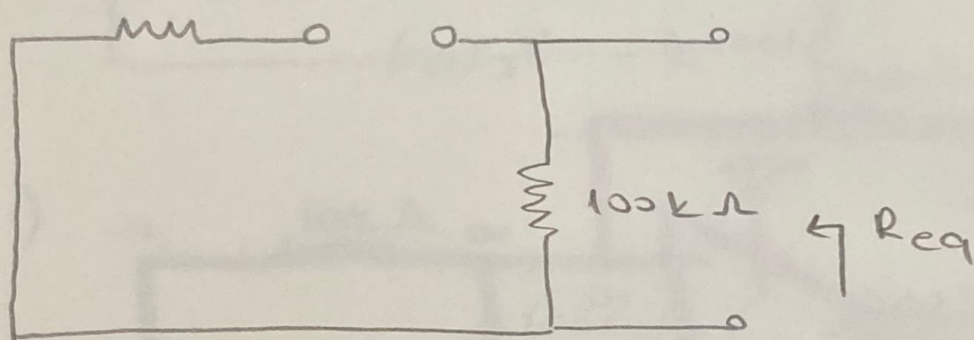
At $t=\infty$



$$V_c(\infty) = 0V$$

$$\begin{aligned}
 a.) \quad V_c(t) &= V_c(\infty) (1 - e^{-t/\tau}) + V_c(0) e^{-t/\tau} \\
 &= -50 e^{-t/\tau}
 \end{aligned}$$

$$\tau = R_{eq} C$$



$$R_{eq} = 100k\Omega$$

$$\tau = 1000000 \times 10^{-6}$$

$$\tau = 0.1 \text{ sec}$$

$$V_c(t) = -50 e^{-t/0.1}$$

$$V_c(t) = -50 e^{-10t} \text{ V for } t \geq 0$$

$$b.) \quad i_c(t) = \frac{C dV_c(t)}{dt} \rightarrow 10^{-6} \times \frac{d(-50 e^{-10t})}{dt}$$

$$= 10^{-6} \times 500 e^{-10t}$$

$$i_c(t) = 0.5 e^{-10t} \text{ mA for } t \geq 0$$